

File Edit Selection

C programming

exp5.c exp_5.c exp1.c exp_8.c exp2.c exp_8.c exp3.c exp_8.c exp1.c exp_7.c exp2.c exp_7.c exp3.c exp_7.c exp1.c exp_8.c

```
exp_6.c> C exp1.c> ...
1 #include <stdio.h>
2
3 // Structure to represent a complex number
4 struct Complex {
5     float real;
6     float imag;
7 };
8
9 // Function to read a complex number
10 struct Complex readComplex() {
11     struct Complex num;
12     printf("Enter the real part: ");
13     scanf("%f", &num.real);
14     printf("Enter the imaginary part: ");
15     scanf("%f", &num.imag);
16     return num;
17 }
18
19 // Function to display a complex number
20 void writeComplex(struct Complex num) {
21     printf("%f + %fi", num.real, num.imag);
22 }
23
24 // Function to add two complex numbers
25 struct Complex addComplex(struct Complex n1, struct Complex n2) {
26     struct Complex result;
27     result.real = n1.real + n2.real;
28     result.imag = n1.imag + n2.imag;
29     return result;
30 }
31
32 // Function to subtract two complex numbers
33 struct Complex subtractComplex(struct Complex n1, struct Complex n2) {
34     struct Complex result;
35     result.real = n1.real - n2.real;
36     result.imag = n1.imag - n2.imag;
37     return result;
38 }
39
40 int main() {
41     struct Complex num1, num2, sum, difference;
42
43     printf("Enter the first complex number:\n");
44     num1 = readComplex();
45
46     printf("\nEnter the second complex number:\n");
47     num2 = readComplex();
48
49     sum = addComplex(num1, num2);
50     difference = subtractComplex(num1, num2);
51
52     printf("\nThe first complex number is: ");
53     writeComplex(num1);
54
55     printf("\nThe second complex number is: ");
56     writeComplex(num2);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

C/C++ Compile Run + - - - - -

```
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output> & .\exp1.exe
Enter the first complex number:
Enter the real part: 4
Enter the imaginary part: 3

Enter the second complex number:
Enter the real part: 1
Enter the imaginary part: 2

The first complex number is: 4.0 + 3.0i
The second complex number is: 1.0 + 2.0i

The sum of the two complex numbers is: 5.0 + 5.0i
The difference of the two complex numbers is: 3.0 + 1.0i
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output>
```

Pointer

Q. Declare diff types of pointers (int, float, char) and initialize them with the addresses of variables. Print the value of both the pointers and the variables they point to.

```
#include <stdio.h>
```

```
int main() {
```

```
    int a = 10
```

```
    float b = 25.75;
```

```
    char c = 'Z';
```

```
    int *p1;
```

```
    float *p2;
```

```
    char *p3;
```

```
    p1 = &a
```

```
    p2 = &b
```

```
    p3 = &c;
```

```
    printf("- Integer pointer - \n");
```

```
    printf("Value of a: %d \n", a);
```

```
    printf("Address of a: %p \n", &a);
```

```
    printf("Pointer p1 (store address of a): %p \n", p1);
```

```
    printf("Value pointed by p1: %d \n \n", *p1);
```

```
    printf("- Float pointer - \n");
```

```
    printf("Value of b: %f \n", b);
```

```
    printf("Address of b: %p \n", &b);
```

```
    printf("Pointer p2 (store address of b): %p \n", p2);
```

```
    printf("Value pointed by p2: %f \n \n", *p2);
```

```
    printf("- Character pointer - \n");
```

```
    printf("Value of c: %c \n", c);
```

```
    printf("Address of c: %p \n", &c);
```

```
    printf("Pointer p3 (store address of c): %p \n", p3);
```

```
    printf("Value pointed by p3: %c \n", *p3);
```

```
    return 0;
```

```
}
```



```

exp_7.c > C exp3.c > main()
1  #include <stdio.h>
2
3  void modifyValues(int *p1, float *p2, char *p3) {
4      *p1 = *p1 + 5;
5      *p2 = *p2 + 10;
6      *p3 = 'A';
7  }
8
9  int main() {
10     int a = 10;
11     float b = 20.5;
12     char c = 'Z';
13
14     printf("The values of the variables before calling the function are:\n");
15     printf("The value of integer variable a is: %d\n", a);
16     printf("The value of float variable b is: %.2f\n", b);
17     printf("The value of character variable c is: %c\n", c);
18
19     modifyValues(&a, &b, &c);
20
21     printf("\nThe values of the variables after calling the function are:\n");
22     printf("The modified value of integer variable a is: %d\n", a);
23     printf("The modified value of float variable b is: %.2f\n", b);
24     printf("The modified value of character variable c is: %c\n", c);
25
26     return 0;
27 }
28

```

```

PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output> & .\exp3.exe

```

The values of the variables before calling the function are:
The value of integer variable a is: 10
The value of float variable b is: 20.50
The value of character variable c is: Z

The values of the variables after calling the function are:
The modified value of integer variable a is: 15
The modified value of float variable b is: 30.50
The modified value of character variable c is: A
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_7.c\output>

Q. Perform pointer arithmetic (inc and decrement) on pointers of different data types. Observe how the memory address changes and the effect on data access

```
#include <stdio.h>
```

```
int main() {
```

```
    int a = 10;
```

```
    float b = 20.5;
```

```
    char c = 'X';
```

```
    int * p1 = &a;
```

```
    float * p2 = &b;
```

```
    char * p3 = &c;
```

```
    printf("Before increment : \n");
```

```
    printf("Address in p1 (int*) = %p\n", p1);
```

```
    printf("Address in p2 (float*) = %p\n", p2);
```

```
    printf("Address in p3 (char*) = %p\n", p3);
```

```
    p1++;
```

```
    p2++;
```

```
    p3++;
```

```
    printf("After increment : \n");
```

```
    printf("Address in p1 (int*) = %p\n", p1);
```

```
    printf("Address in p2 (float*) = %p\n", p2);
```

```
    printf("Address in p3 (char*) = %p\n", p3);
```

```
    p1--;
```

```
    p2--;
```

```
    p3--;
```

```
    printf("After decrement : \n");
```

```
    printf("Address in p1 (int*) = %p\n", p1);
```

```
    printf("Address in p2 (float*) = %p\n", p2);
```

```
    printf("Address in p3 (char*) = %p\n", p3);
```

```
    return 0;
```



```

exp_6.c> C exp_6.c ...
1 #include <stdio.h>
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3 // Structure to represent a complex number
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6     float imag;
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8
9 // Function to read a complex number
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12     printf("Enter the real part: ");
13     scanf("%f", &num.real);
14     printf("Enter the imaginary part: ");
15     scanf("%f", &num.imag);
16     return num;
17 }
18
19 // Function to display a complex number
20 void writeComplex(struct Complex num) {
21     printf("%.1f + %.1fi", num.real, num.imag);
22 }
23
24 // Function to add two complex numbers
25 struct Complex addComplex(struct Complex n1, struct Complex n2) {
26     struct Complex result;
27     result.real = n1.real + n2.real;
28     result.imag = n1.imag + n2.imag;
29     return result;
30 }
31
32 // Function to subtract two complex numbers
33 struct Complex subtractComplex(struct Complex n1, struct Complex n2) {
34     struct Complex result;
35     result.real = n1.real - n2.real;
36     result.imag = n1.imag - n2.imag;
37     return result;
38 }
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40 int main() {
41     struct Complex num1, num2, sum, difference;
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43     printf("Enter the first complex number:\n");
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47     num2 = readComplex();
48
49     sum = addComplex(num1, num2);
50     difference = subtractComplex(num1, num2);
51
52     printf("\nThe first complex number is: ");
53     writeComplex(num1);
54
55     printf("\nThe second complex number is: ");
56     writeComplex(num2);
57

```

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```

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Enter the first complex number:
Enter the real part: 4
Enter the imaginary part: 3

Enter the second complex number:
Enter the real part: 1
Enter the imaginary part: 2

The first complex number is: 4.0 + 3.0i
The second complex number is: 1.0 + 2.0i

The sum of the two complex numbers is: 5.0 + 5.0i
The difference of the two complex numbers is: 3.0 + 1.0i
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output>

```

Q. Write a function that accepts pointers as parameters. Pass variable by reference using pointers and modify their val within the function.

#include <stdio.h>

void modifyValues(int *x, float *y, char *z) {

*x = *x + 5;

*y = *y * 2;

*z = *z + 1;

}

int main() {

int a = 10;

float b = 20.5;

char c = 'A';

printf("Before function call: \n");

printf("a = %d \n", a);

printf("b = %.2f \n", b);

printf("c = %c \n", c);

modifyValues(&a, &b, &c);

printf("\n After function call: \n");

printf("a = %d \n", a);

printf("b = %.2f \n", b);

printf("c = %c \n", c);

return 0;

Output: Before function call: a = 10, b = 20.5, c = A After function call: a = 15, b = 41, c = A